

SAINTGITS COLLEGE OF ENGINEERING (AUTONOMOUS), KOTTAYAM

First Series Examinations, August 2025

24BSE2113-D Partial Differential Equations, Transforms and Optimization Methods

Duration: 90 minutes

Maximum marks: 50

Part A (Answer all questions)

1. Find the partial differential equation of $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$. (5 marks)
2. (a) Form a PDE by elimination of the arbitrary constants a and b from $z = xy + y\sqrt{x^2 + y^2} + b$. (3 marks)
(b) Solve using Lagrange method, $px - qy = y^2 - x^2$. (7 marks)
3. Using method of separation of variable, solve $u_{xy} - u = 0$. (10 marks)

Part B Answer all questions.

4. State all basic assumptions in the solution of one dimensional wave equation of second order and write all possible solutions under these assumptions. (5 marks)
5. A tightly stretched string of length l with fixed ends is initially in equilibrium position. Motion is governed by $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ and is started by displacing the string into the form of the curve $y = kx(l - x)$, $0 < x < l$, from which it is released at time $t = 0$. Find the displacement of any point at a distance x from one end at time t . (10 marks)
6. Derive the solutions of one dimensional heat equation using the method of separation of variables. (10 marks)